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GlucoTwin

A personal digital twin that sees a Type 2 Diabetes glucose spike two hours before it happens

Be Positive · IIM Kozhikode
Digital Twin Challenge 2026 · Happiest Health · Reimagining & Reforming Healthcare in India Summit

India's diabetes care is still reactive

101 M

Indians living with diabetes (ICMR-INDIAB, Lancet D&E 2023)

136 M

more with prediabetes, the pipeline of tomorrow's patients

~3 / yr

typical HbA1c checks: a 90-day average that hides daily spikes

The gap: post-meal glucose spikes drive vascular damage and HbA1c, but they happen between clinic visits. CGMs and smartwatches now generate the data, yet they only show what already happened; the patient learns about a spike after it peaks.

What a clinician needs: a warning early enough to act (portion, walk, medication), personalised to this patient's physiology, explained in plain terms, and with few false alarms.

Why a digital twin, not just another ML model

1 Pure ML

Learns correlations from data. Strong with lots of data, but a black box, cannot answer "what if she walks after dinner?" and degrades for unusual patients.

2 Pure physiology

Equations of glucose and insulin. Explainable and can simulate interventions, but generic parameters fit real patients poorly and ignore context such as sleep and stress.

3 GlucoTwin = hybrid

A physiological model personalised to each patient and kept in sync with live data, whose forecasts feed an ML risk layer. Accurate, explainable, and able to run counterfactuals.

Design principle: *the twin is the patient's virtual copy; the ML layer is the alarm that listens to it.*

Scope: one condition, one adverse event, one decision

Use case

- Condition: Type 2 Diabetes and prediabetes
- Patient: adult with a CGM + smartwatch, managed in an outpatient / clinic model
- User: treating doctor or diabetes educator; nudges reach the patient
- Decision supported: intervene before the meal or right after it

Adverse event predicted

CGM glucose > 180 mg/dL within the next 120 minutes

- 180 mg/dL = upper limit of the international time-in-range consensus
- Predicted only while the patient is still ≤ 180, so every alert is a genuine early warning
- Risk refreshed every 15 minutes

Two data streams, fully synthetic and privacy-safe

E Static / historical: EHR

- Demographics, BMI, waist, BP
- ICD-10 diagnoses (E11.9, R73.03, I10)
- LOINC labs: HbA1c, fasting glucose & insulin, lipids, eGFR
- Medications: metformin, SU, DPP-4i, insulin
- Genetic markers: TCF7L2, KCNQ1, FTO

W Dynamic / real-time: wearables

- CGM glucose every 5 min
- Heart rate and HRV (RMSSD)
- Step count
- Sleep stages: wake / light / deep / REM
- Patient's own meal log: incomplete & noisy

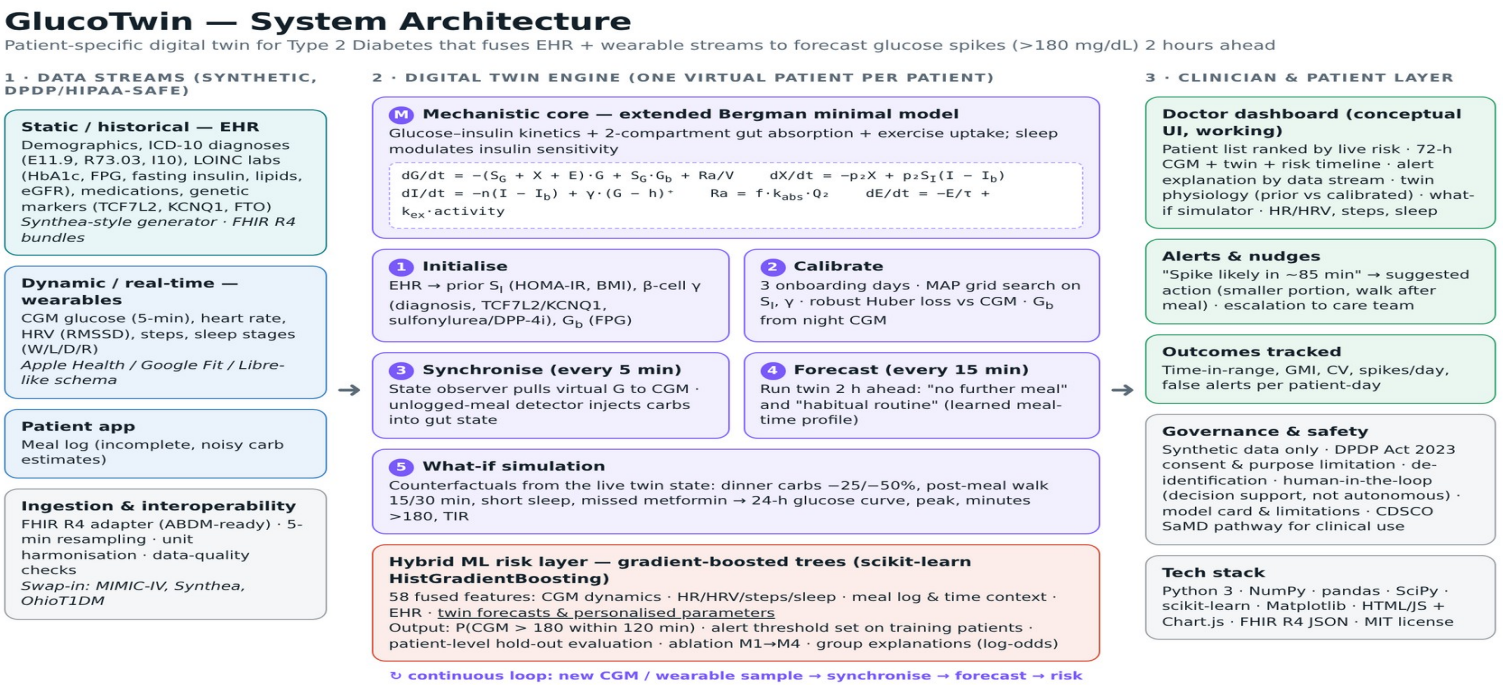
100
synthetic patients

14 days
per patient, 5-min data

403 k
CGM + wearable rows

0
real patient records

Architecture



How each twin is born and kept alive

1

Initialise

EHR sets prior physiology: insulin sensitivity from HOMA-IR & BMI; β-cell response from diagnosis, genotype, drugs; basal glucose from FPG

2

Calibrate

First 3 days: MAP search over insulin sensitivity and β-cell response against CGM, robust to unlogged meals

3

Synchronise

Every 5 min a state observer pulls the virtual glucose onto the CGM and detects meals the patient forgot to log

4

Forecast

Every 15 min the twin runs 2 h ahead: "no further meal" and "habitual routine" from its learned meal-time profile

5

What-if

Doctor tests interventions on the live state: smaller dinner, walk, short sleep, missed metformin

Sleep enters the physiology: short or poor sleep lowers next-day insulin sensitivity; walking adds insulin-independent glucose uptake; stress raises hepatic glucose output.

The twin learns each patient's physiology

−30%

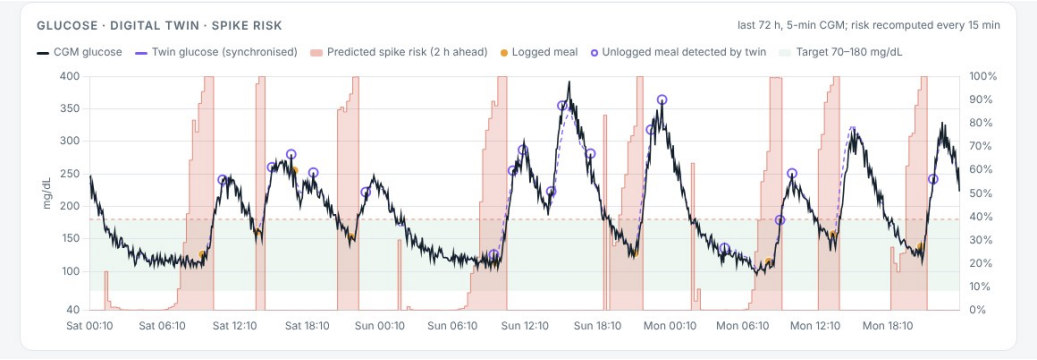
error in recovered insulin sensitivity vs EHR-only prior (log-scale, against hidden ground truth)

51 → 46

mg/dL open-loop RMSE over 3 days after calibration (no data assimilation)

6.8

mg/dL mean absolute error once synchronised to live CGM every 5 min

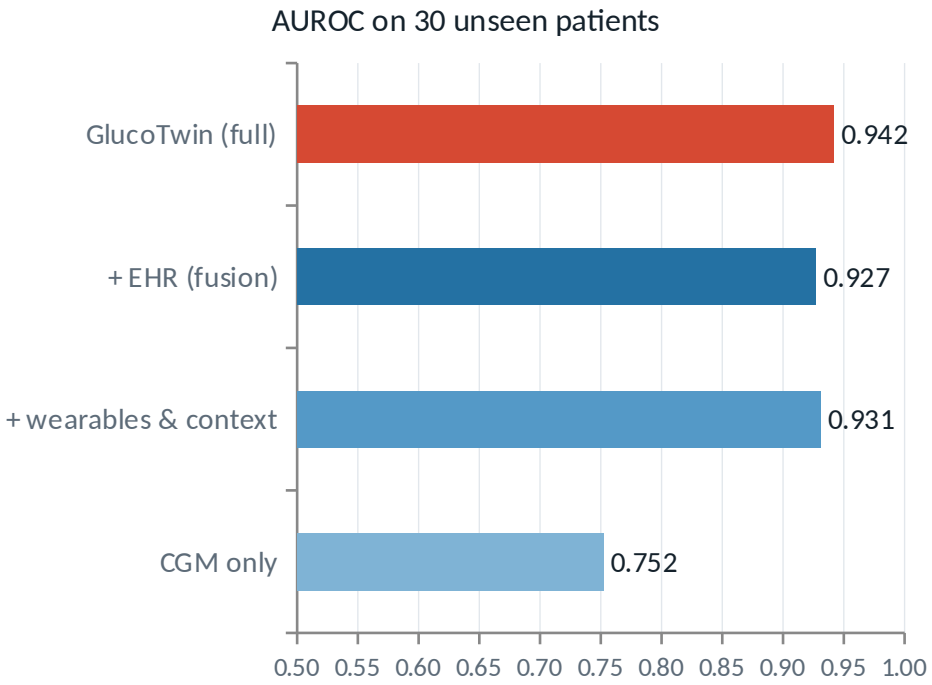


Purple dashed line = twin glucose tracking the patient's CGM (black).

Hollow purple circles = meals the patient never logged, detected by the twin from the glucose trajectory.

Red area = live spike risk; it rises before each excursion.

Every stream adds signal; the twin adds the most



Risk model

- Gradient-boosted trees (scikit-learn), 58 features in 5 streams
- Patient-level split: 70 train / 30 test patients, never mixed
- 56 k training, 23 k test samples; 31% spike prevalence
- Twin features: 2-h forecast peaks, meal probability, personalised SI & γ, unlogged carbs, residual

Clinically: earlier warnings, far fewer false alarms

0.942

AUROC on unseen patients (AUPRC 0.888)

77%

of spike episodes flagged before onset

85 min

median warning before the spike

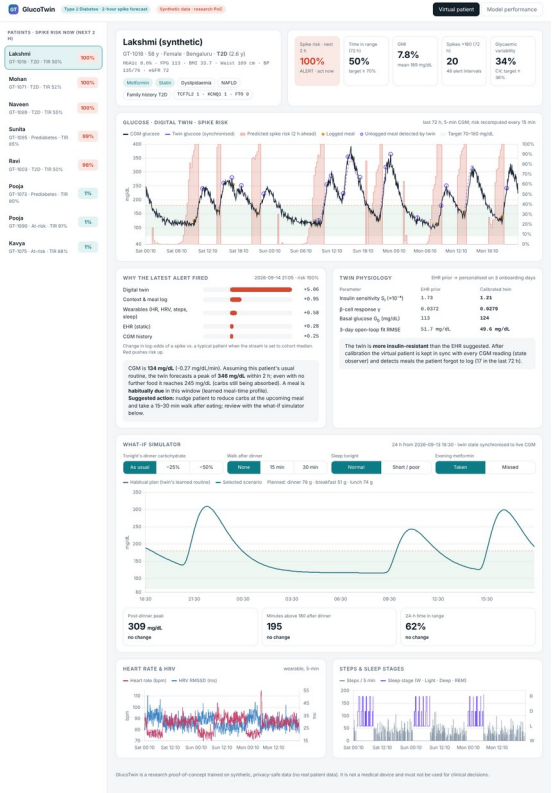
4.6×

fewer false alerts than CGM-only (0.79 vs 3.61 per patient-day)

Model	AUROC	Episodes caught	Median lead	False alerts / pt-day
Rule: CGM > 140 and rising	0.546	54%	10 min	0.40
CGM-only ML	0.752	87%	90 min	3.61
+ wearables, context, EHR	0.927	78%	75 min	0.83
GlucoTwin (full)	**0.942**	**77%**	**85 min**	**0.79**

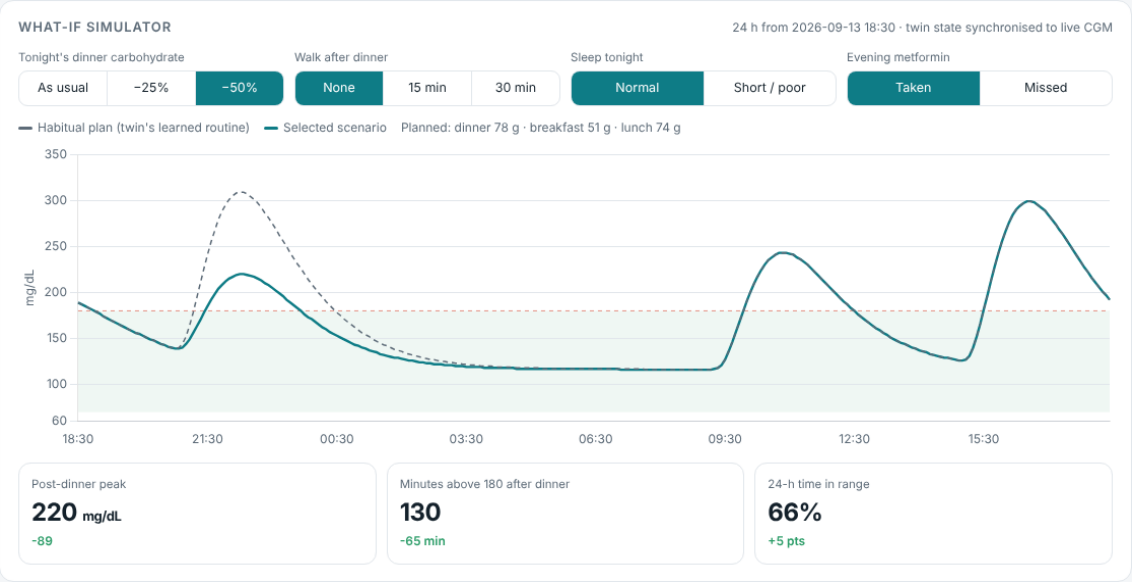
Alert threshold fixed on training patients only (80% sample sensitivity). Calibrated probabilities; consistent across T2D (AUROC 0.94), prediabetes (0.94) and at-risk (0.93).

The doctor's view of the virtual patient



1. Patient list ranked by live spike risk
2. 72-h timeline: CGM, twin glucose, meals (logged & detected), risk
3. "Why this alert": contribution of each data stream + plain-language reasons & suggested action
4. Twin physiology: EHR prior vs personalised parameters
5. What-if simulator, then HR / HRV, steps and sleep

What-if: test the intervention on the twin first



−89 mg/dL

post-dinner peak when tonight's dinner carbs are halved (309 → 220)

−65 min

time above 180 after dinner, for the same change

Also simulated: 15/30-min post-dinner walk, a short night's sleep (lower next-day insulin sensitivity), and a missed metformin dose.

From proof of concept to a Happiest Health pilot

0–3 months

Retrospective validation

Re-run on de-identified CGM + EHR data (OhioT1DM, MIMIC-IV, partner clinic) under DPDP consent; recalibrate population priors for Indian cohorts

3–9 months

Clinic pilot

100–200 consenting patients in Happiest Health clinics; dashboard for doctors and educators; app nudges; measure time-in-range and alert fatigue

9–18 months

Scale & regulate

ABDM / FHIR integration, more conditions on the same twin core (hypertension, CVD risk); CDSCO SaMD pathway

Value: each 10-point gain in time-in-range is linked to lower risk of complications; spikes are prevented between visits, care teams prioritise by risk, and patients get personal, quantified advice.

Limitations and responsible AI

Honest limitations

- Synthetic data: results show the method works, not clinical accuracy
- Minimal model simplifies physiology (no glucagon, no meal composition beyond carbs)
- Meal timing habits are a strong signal; irregular eaters will be harder
- Needs real-world validation before any clinical use

Safeguards built in

- Privacy by design: no real data; DPDP consent & purpose limitation for pilots
- Human in the loop: decision support for clinicians, never autonomous dosing
- Explainable alerts and a physiological model doctors can inspect
- Subgroup checks, calibration, open-source code (MIT)

Proactive care, one twin at a time

EHR + wearables → calibrated, synchronised virtual patient → 85-min warning with 4.6× fewer false alarms → personal what-if advice

Run it: python run_all.py → open dashboard/index.html
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